MARZUQUE MASHRUR FARIZ

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EDUCATION

University of Texas - Arlington **August 2022 - May 2026**
Bachelor's, Computer Science *GPA: 3.93*

PROFESSIONAL EXPERIENCE

Meezan Auto Group LLC **August 2026 - Present**
Automation & AI Engineer *Dallas, TX, USA*

- Developed machine learning workflows and AI-powered features using historical vehicle, pricing, repair, and sales data to identify purchasing opportunities and raise vehicle value estimation accuracy by 15%.
- Designed and built an internal vehicle-tracking system to manage inventory across the auction, acquisition, repair, listing, and sales lifecycle, reducing vehicle life-cycle track times by 37%.
- Automated manual document-processing workflows using agentic AI for purchase and sales records, including digital signing, file organization, uploading, and record management.
- Built dashboards, reports, and internal tools to analyze financial data, expenses, profit margins, inventory, and sales performance for operational decision-making.
- Maintain and improve dealership web and digital systems while developing invoices, reports, and standardized documentation to support day-to-day business operations.

PROJECTS & OUTSIDE EXPERIENCE

Secure Code Generation with LLMs **April 2025 - August 2026**
Honors Undergraduate Researcher

- Developed a prompt-level security evaluation framework on LangChain that predicts code risk from a prompt's natural-language structure and the specifications it elicits.
- Expanded available data by more than 50% through an expanding dataset that connects prompts, task specifications, and vulnerability outcomes to support systematic evaluation of prompt-driven security.
- Implemented a security-testing pipeline using Bandit and CodeQL, and organized results into reusable vulnerability/mode-of-failure categories.
- Analyzed linguistic, structural, and semantic prompt features that correlate with insecure outputs, aiming for token-efficient, real-time prompt checking.
- Synthesized 15+ research papers to ground threat modeling, define evaluation criteria, and guide dataset design and experimental methodology.

Autograd-Free Neural Network Trainer **January 2026 - July 2026**

- Implemented neural networks end-to-end: forward pass, backpropagation, gradient descent, activations, and loss functions, with and without ML libraries.
- Validated learning behavior through controlled experiments to understand training dynamics and optimization failure modes.
- Strengthened fluency in the mathematical foundations of deep learning (gradients, stability, generalization).

Graduate Compilers - LLVM-Compatible Compiler **August 2025 - December 2025**

- Implemented a full compiler pipeline, including lexical analysis, recursive-descent parsing, AST construction, semantic checks, and LLVM IR generation.
- Designed intermediate representations and symbol-table structures that strengthened understanding of language semantics and program analysis.
- Gained experience with low-level code generation and compiler correctness, system design, foundational for programming languages research.

SKILLS

Programming Languages: C/C++, Python, Java, R, SQL, Perl, Go
Technical Skills: LLM, RAG, REST APIs, DevOps, Version Control, LangChain, LangGraph